Daniele Zago

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Date of birth: 9 May 1996

Research-oriented data scientist and optimization engineer bridging **statistics**, **machine learning**, and **mathematical optimization**. I design and implement end-to-end solutions spanning stochastic optimization algorithms, ML models, and production software on enterprise stacks, with a track record in both academic research and industrial deployment.

Work Experience

Data Scientist Oct 2024 – Present
Optit S.r.l. *Bologna, Italy*

Technologies: **Python, Java, Hexaly (LocalSolver), Xpress Mosel, Gradle, Docker, scikit-learn, MLFlow**

- Research and development of solutions to the **periodic vehicle routing problem** with clustered routes for multi-day strategic fleet management, including front-end development for interactive visualization and exploration
- Improved **minimum cost flow** approaches for crossdock facility selection when serving customers from warehouses with potential crossdock drop-off of goods
- Developed **forecasting and anomaly detection** ML-based pipelines for power plant energy consumption

Statistical consultant Jul 2023 – Oct 2023
Expin S.r.l. *Padua, Italy*

Technologies: **R, C++**

Developed a sequential monitoring system for anomaly detection in vibrating string strain gauges, and a novel algorithm for optimal alarm threshold selection using stochastic optimization techniques.

Teaching assistant Oct 2022 – Dec 2022
Department of Developmental Psychology and Socialisation, University of Padua *Padua, Italy*

Lectures on factor models and analysis of questionnaires, laboratory sessions on programming.

Academic tutor Sep 2017 – Jan 2019
Department of Statistical Sciences, University of Padua *Padua, Italy*

Provided calculus lessons and interactive workshops for undergraduate students, in preparation for the exams.

Education

University of Padua Padua, Italy
Ph.D. in Statistical Sciences *2021 – 2024*

- Advisor: prof. Giovanna Capizzi; Co-advisor: prof. Peihua Qiu
- Research topic(s): **online outlier detection** and **stochastic optimization**

University of Florida Gainesville, FL, USA
Visiting research scholar, supervisor: Prof. Peihua Qiu *Jan 2023 – Dec 2023*

INFN Bertinoro, Italy
Thirteenth INFN International School on Efficient Scientific Computing *Oct 2022*

- Efficient C++ programming, GPU programming with CUDA

University of Padua Padua, Italy
M.Sc. in Statistical Sciences *2019 – 2021*

- Final grade: **110/110 cum Laude**, GPA: **29.5/30**
- Thesis topic(s): **Bayesian nonparametric mixture models**

University of Perugia Perugia, Italy
Summer school in Mathematics (functional analysis and mathematical statistics) *Jul 2020*

University of Padua Padua, Italy
B.Sc. in Statistics for Technology and Sciences *2016 – 2019*

- Final grade: **110/110 cum Laude**, GPA: **29.2/30**
- Thesis topic(s): **applied Bayesian modelling**

Skills

Expertise	Mathematical and stochastic optimization, constrained optimization, statistical process monitoring, machine learning, anomaly detection, statistical modelling
Programming	R, Julia, Python, C++, Rust, Java, TypeScript, SQL, Stan, bash, SAS
Tools	git, MLFlow, Hexaly, Gradle, Docker, Microsoft Office
Languages	Italian (native), English (fluent, C2), German (moderate), Spanish (moderate)

Publications

Journal articles

Zago, D. (2025). StatisticalProcessMonitoring.Jl: A General Framework for Statistical Process Monitoring in Julia. *Journal of Statistical Software* 113, 1–45. doi: 10.18637/jss.v113.i07 *(open-source Julia package)*

Zago, D., Capizzi, G., and Qiu, P. (2025). An Improved Bisection-Type Algorithm for Control Chart Calibration. *Statistics and Computing* 35, 81. doi: 10.1007/s11222-025-10609-7

Zago, D., Tian, Z., Capizzi, G., and Qiu, P. (2025). A General Framework for Monitoring Mixed Data. *Journal of Quality Technology* 57, 282–296. doi: 10.1080/00224065.2025.2512164

Zago, D., Capizzi, G., and Qiu, P. (2024). Optimal Constrained Design of Control Charts Using Stochastic Approximations. *Journal of Quality Technology* 56, 257–275. doi: 10.1080/00224065.2024.2323585

Zago, D., and Capizzi, G. (2024). Alternative Parameter Learning Schemes for Monitoring Process Stability. *Quality Engineering* 36, 560–574. doi: 10.1080/08982112.2023.2253891

Awards

2025	ENBIS Knowledge Fund, ENBIS 2025 conference	*Piraeus, Greece*
2022	Young Travel Award, ISBA 2022 conference	*Montréal, Canada*
2018	Mille e una Lode Award 2018 *(top 3% of students)*	*University of Padua*
2017	Mille e una Lode Award 2017 *(top 3% of students)*	*University of Padua*

Conference Presentations

Nov 2025 **Invited seminar**. *University of Padua* — *Padua, Italy*
Efficient algorithms for control limit calibration

Sep 2025 **Invited talk**. *ENBIS-25 Conference* — *Piraeus, Greece*
Optimal constrained design of control charts using stochastic approximations

Oct 2023 **Invited talk**. *2023 INFORMS Annual Meeting* — *Phoenix, AZ, USA*
Optimal constrained design of control charts using stochastic approximations

Sep 2022 **Poster presentation**. *Statistical methods and models for complex data* — *Padova, Italy*
Profile monitoring based on adaptive parameter learning

Jun 2022 **Poster presentation**. *2022 ISBA World meeting* — *Montréal, Canada*
Bayesian nonparametric multiscale mixture models via Hilbert-curve partitioning

References

Peihua Qiu
Dean's Professor of Biostatistics
Department of Biostatistics
University of Florida
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Nicola Sartori
Full professor
Department of Statistical Sciences
University of Padua
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